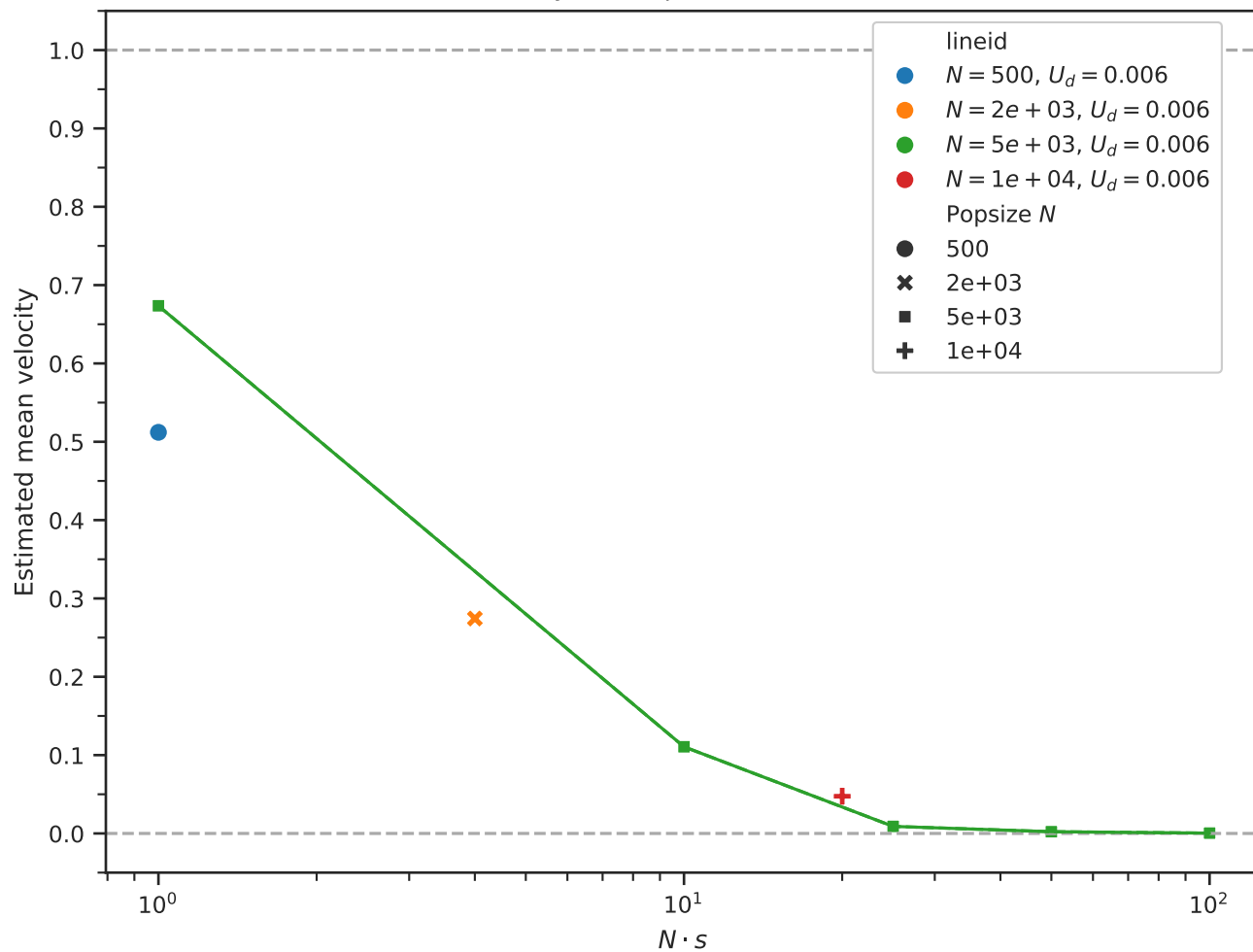
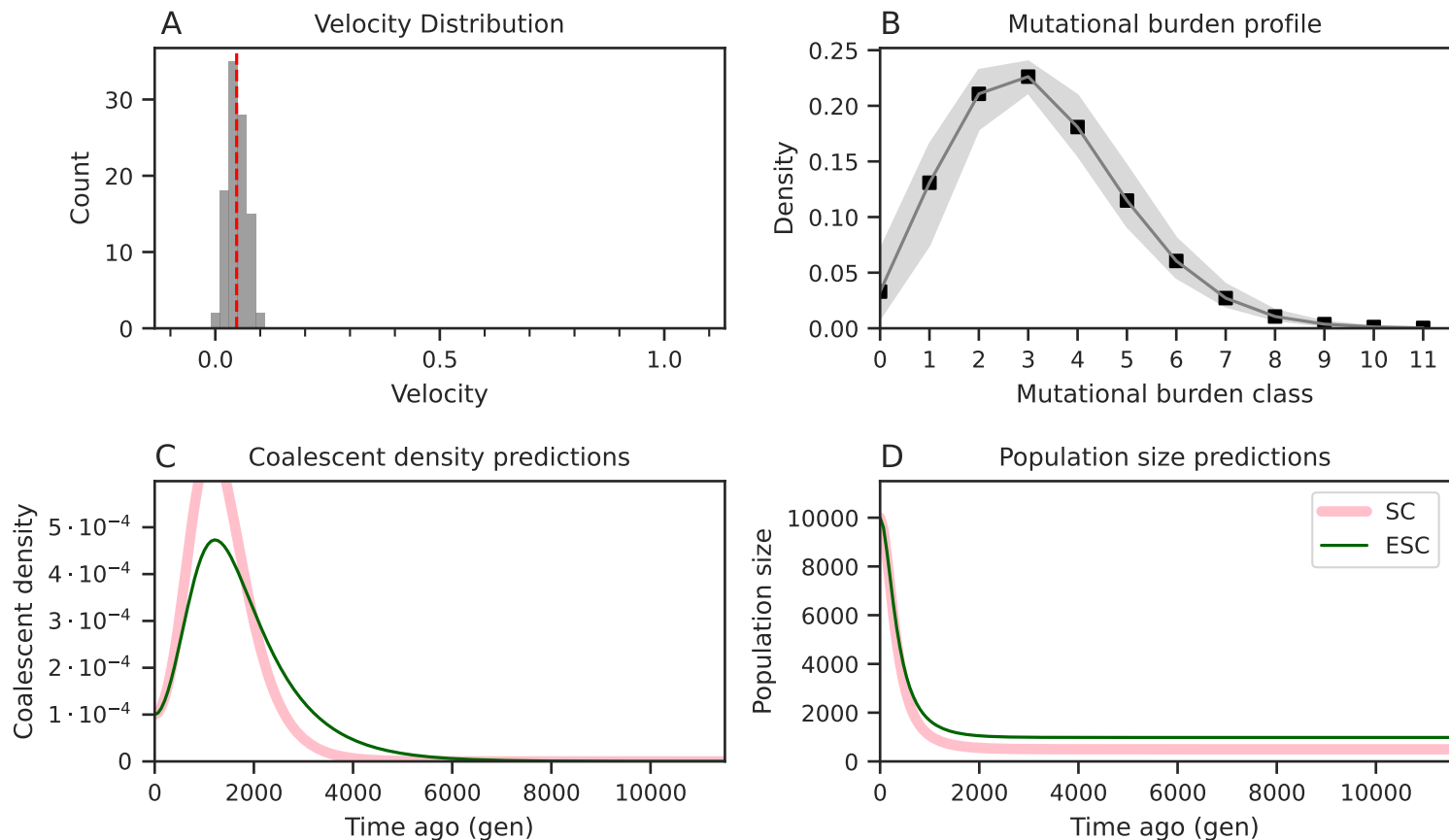


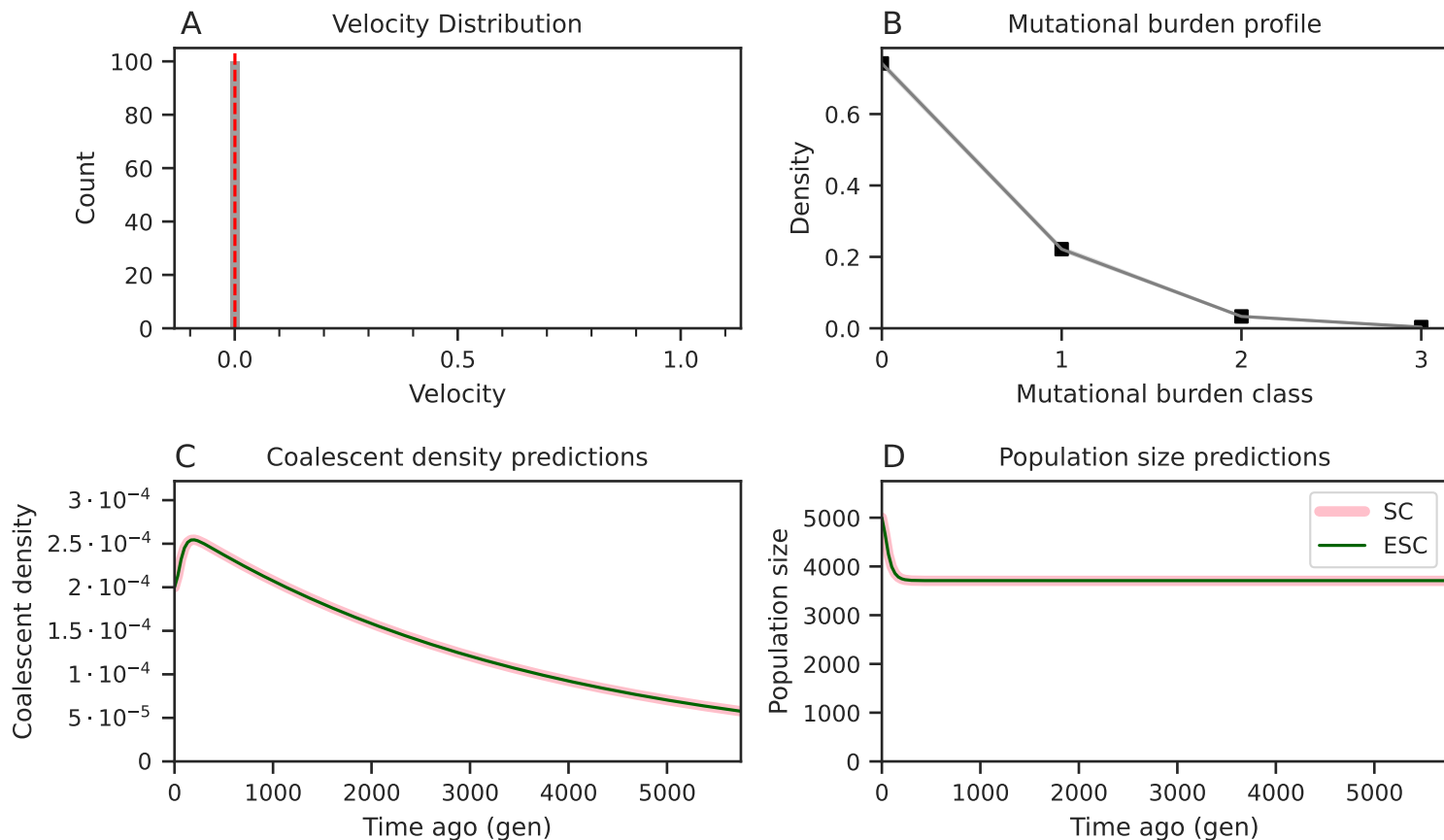
Mean velocity across parameter combinations



$N = 1e + 04$, $s = 0.002$, $U_d = 0.006$, $\phi = 0.996$
Mean Velocity= 0.0475 ± 0.0192 , $n = 100$ simulations

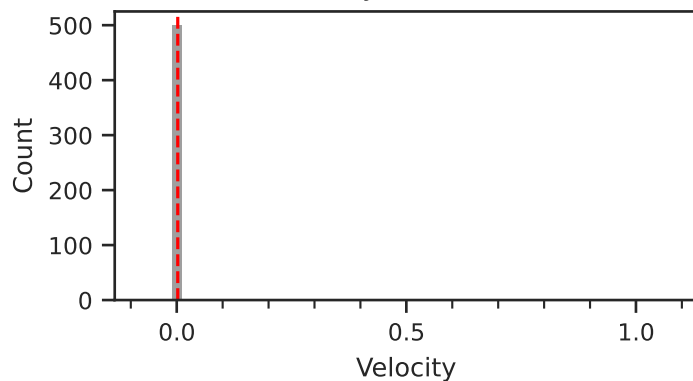


$N = 5e + 03$, $s = 0.02$, $U_d = 0.006$, $\phi = 74.1$
Mean Velocity= 0.0004 ± 0.0005 , $n = 100$ simulations

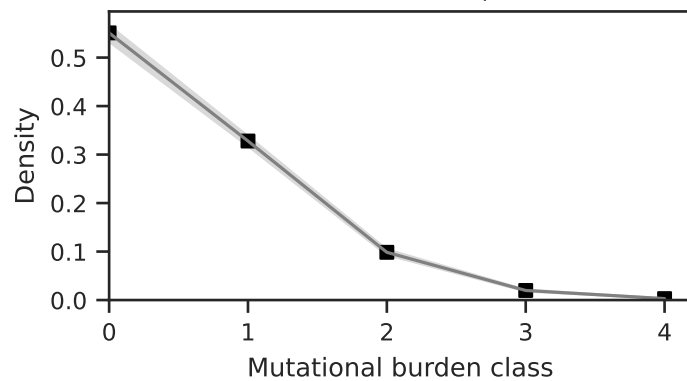


$N = 5e + 03$, $s = 0.01$, $U_d = 0.006$, $\phi = 27.4$
Mean Velocity= 0.0022 ± 0.0014 , $n = 500$ simulations

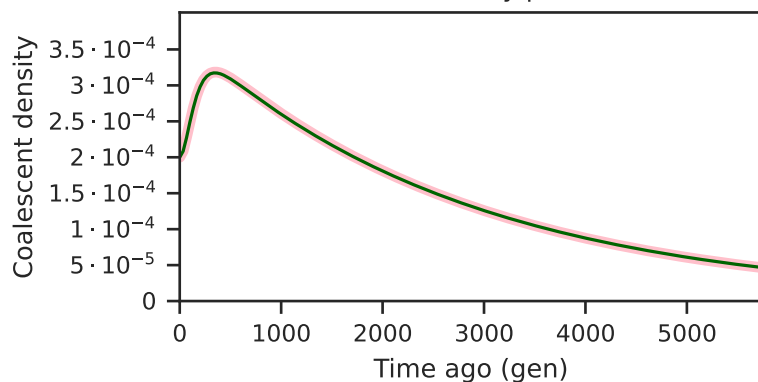
A Velocity Distribution



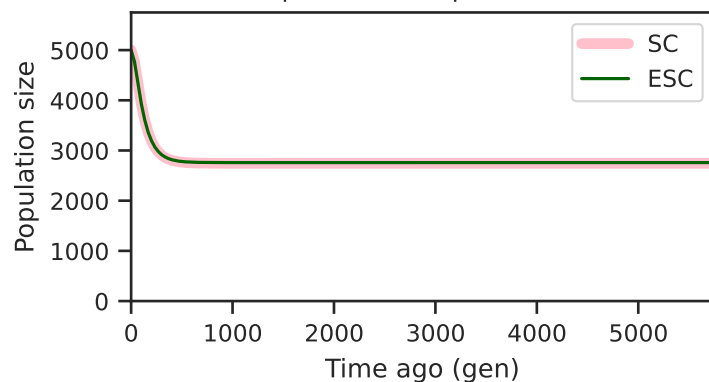
B Mutational burden profile



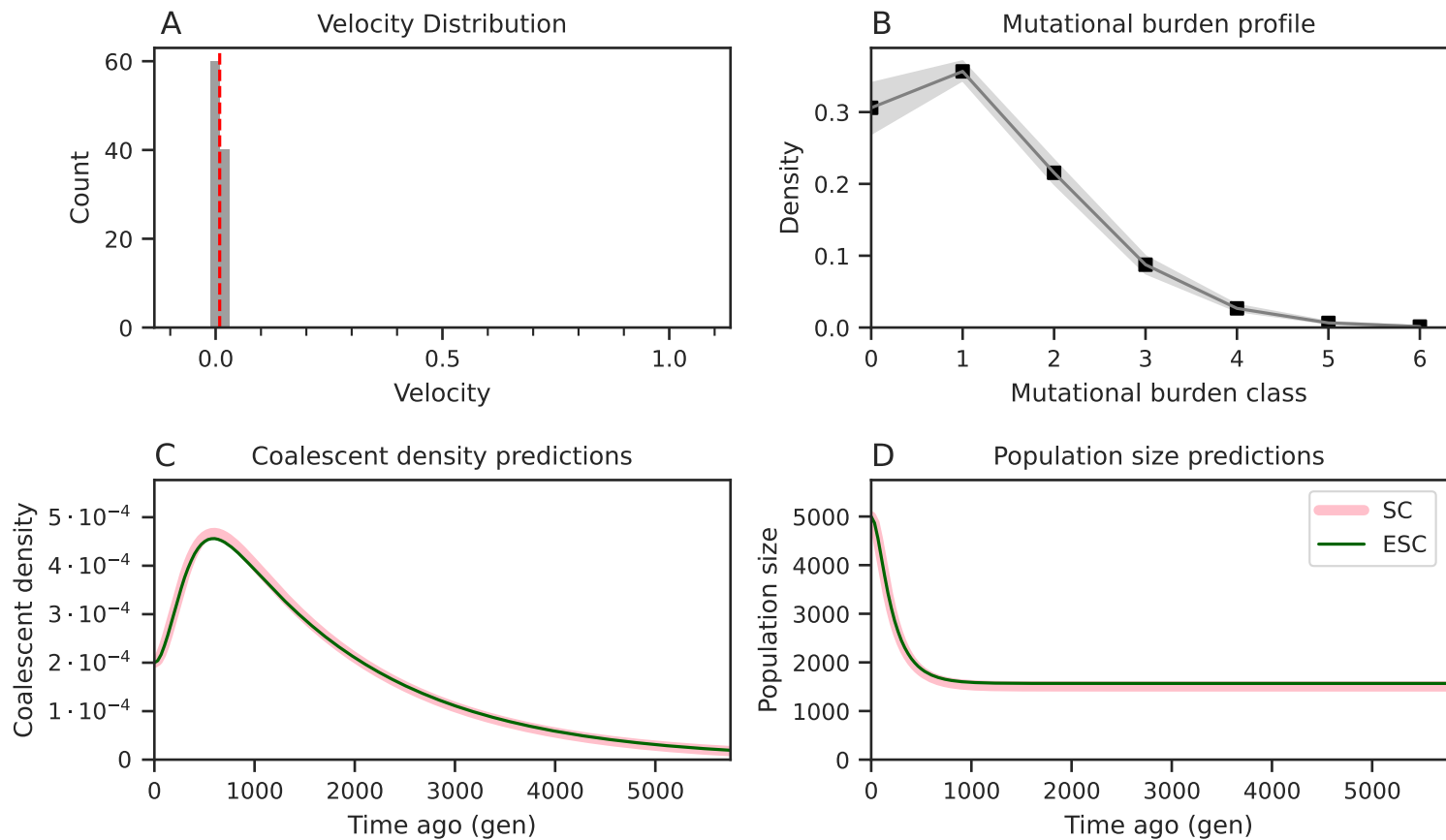
C Coalescent density predictions



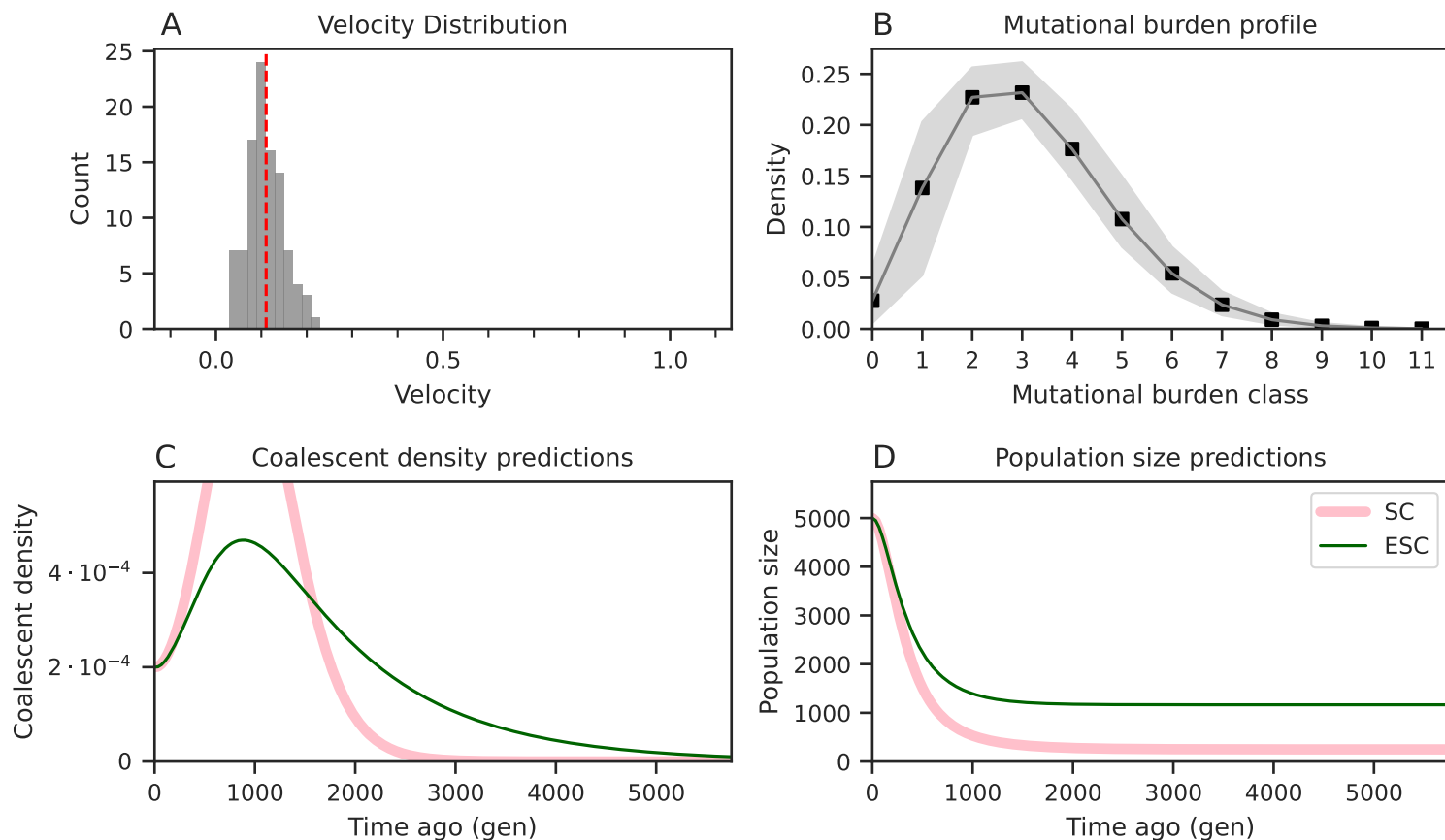
D Population size predictions



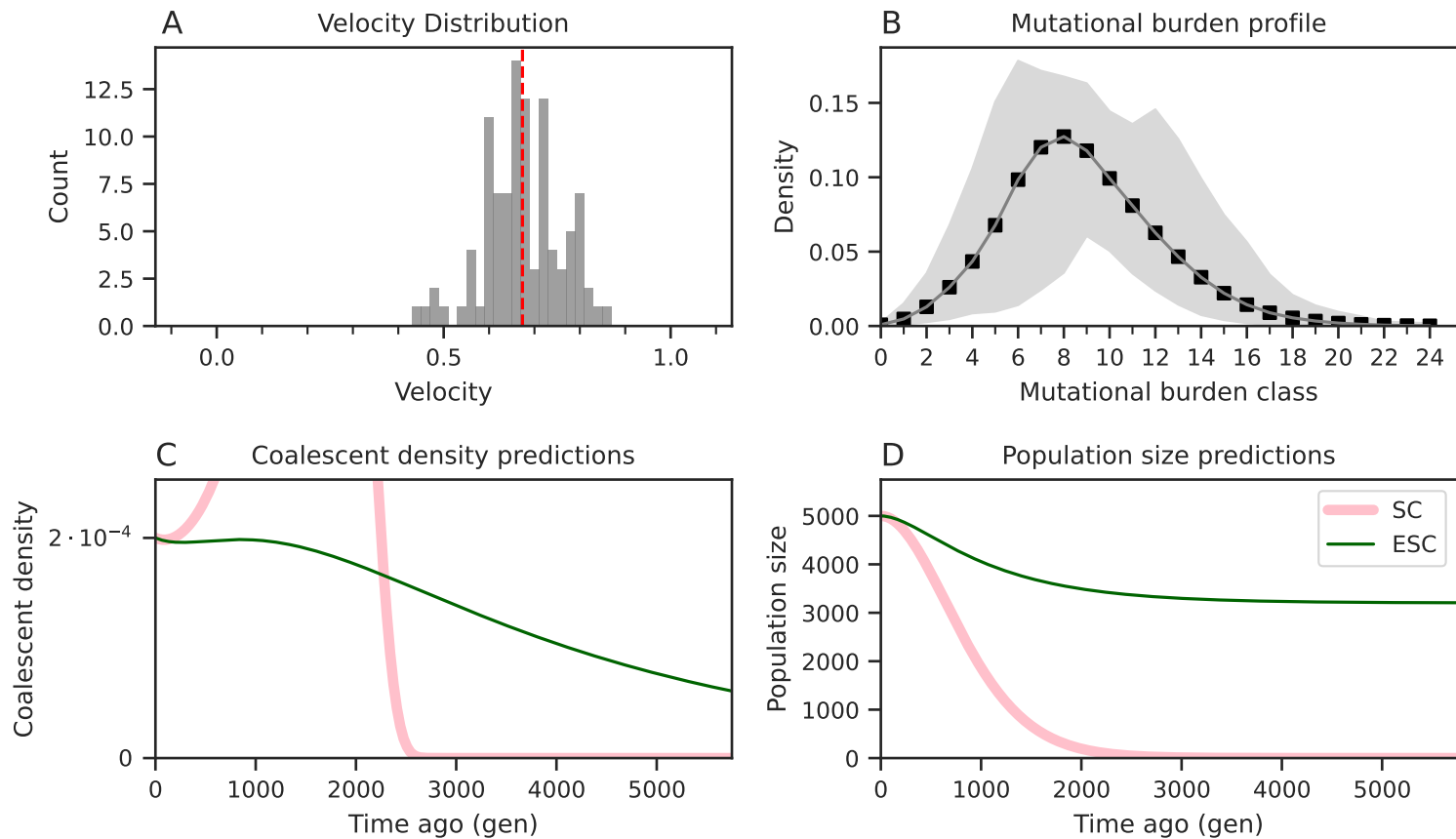
$N = 5e + 03$, $s = 0.005$, $U_d = 0.006$, $\phi = 7.53$
Mean Velocity= 0.0090 ± 0.0056 , $n = 100$ simulations



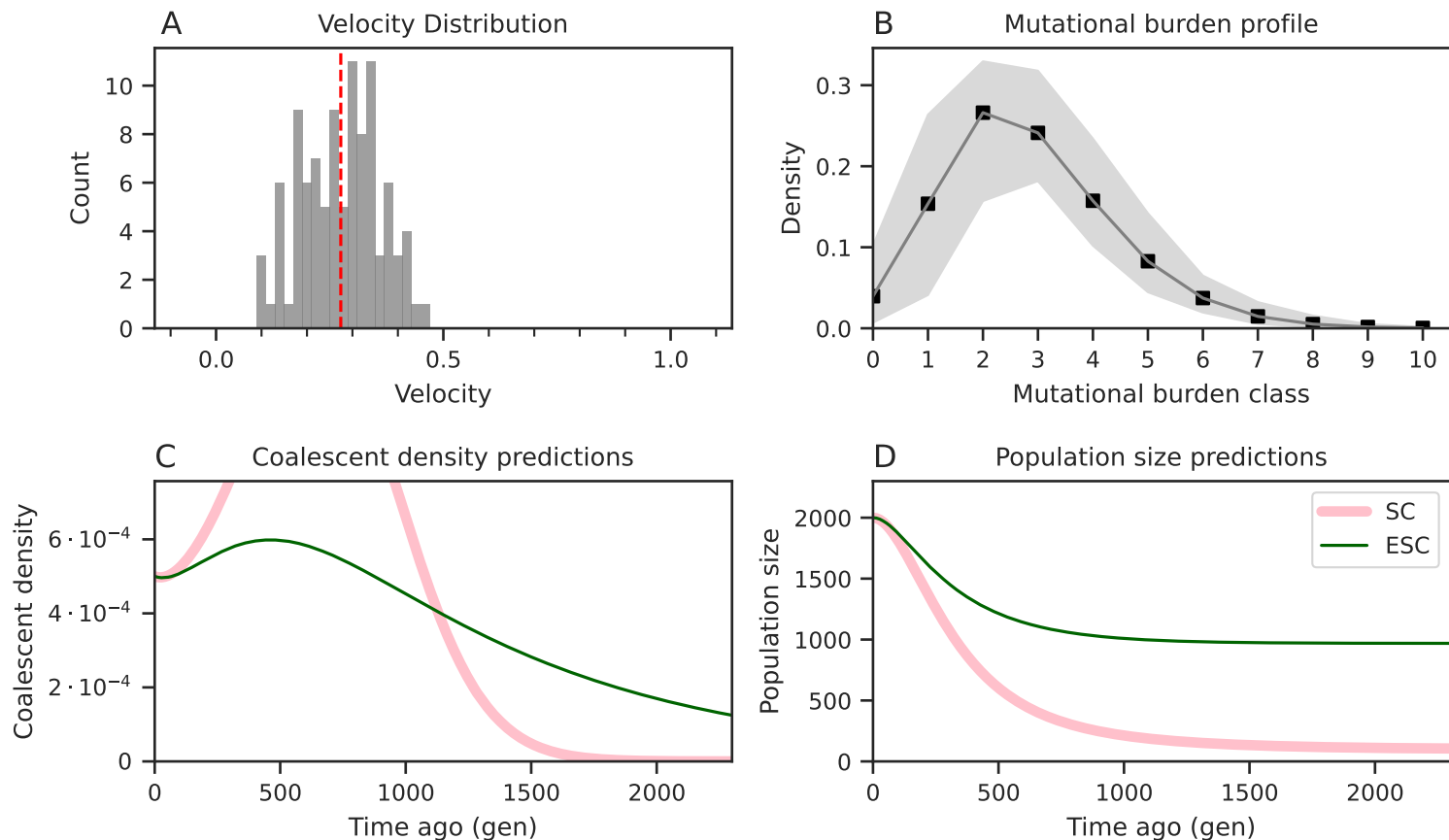
$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\phi = 0.498$
Mean Velocity= 0.1106 ± 0.0399 , $n = 100$ simulations



$N = 5e + 03$, $s = 2 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 9.36 \cdot 10^{-14}$
Mean Velocity= 0.6736 ± 0.0858 , $n = 100$ simulations



$N = 2e + 03$, $s = 0.002$, $U_d = 0.006$, $\phi = 0.199$
Mean Velocity = 0.2744 ± 0.0857 , $n = 100$ simulations



$N = 500, s = 0.002, U_d = 0.006, \phi = 0.0498$
Mean Velocity = 0.5121 ± 0.1834 , $n = 100$ simulations

